

Master Internship Report

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Climate Change vs. Internal Variability of Compound Flood Drivers in Norway: A Risk Perspective

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1 Introduction

From August 7th to 9th 2023, Storm Hans broke century-old rainfall records in eastern Norway, leading to severe flooding and landslides. Homes, roads, railways and bridges were damaged at an estimated cost of 4 billion NOK. More than 10,000 insurance claims were filed and approximately 2,400 people were evacuated, which accounts for the largest evacuation in Norway since the Second World War. This extreme event therefore raises key questions for societal preparedness such as the rarity of Storm Hans and the way society can anticipate and prepare for such unprecedented extremes.

1.1 Challenge in estimating rarity of extreme events

Answering this question is first and foremost difficult because such events are very rare. Traditional risk estimates are derived from observational records with an extent of a few decades, resulting in most extreme events only being represented, if even, by a few samples. Extrapolating beyond the observed range therefore carries large uncertainty, and an event such as storm Hans may appear either unprecedented or unremarkable depending on the record used. This motivates the use of large-ensemble weather and climate model output to enlarge the sample of physically plausible extreme events. These ensembles from global or regional climate models provide many realisations of the same climate as in the actual observational record and are able to simulate sample events that could plausibly have occurred but did not. From a practical standpoint, however, it remains unclear which of these approaches is most appropriate for preparedness planning, and what their respective strengths and limitations are.

1.2 UNSEEN approach

One of these approaches is the UNSEEN method (UNprecedented Simulated Extremes using ENsembles), introduced by Thompson et al. [2017]. UNSEEN treats the individual members of a forecast ensemble as independent, physically consistent realisations of the present climate. By using large ensembles of single-model initial conditions, the UNSEEN method provides a much larger statistically independent sample of realisations than the observational record of the same climate. It therefore reveals physically plausible but so far unobserved extremes and allows adaptation measures to be stress-tested against a maximum credible event rather than against the largest event that has been observed so far [Wiel et al., 2019]. This gives UNSEEN considerable potential as a tool for decision-making under unforeseen hydro-climatic risk.

1.3 Aims of the study

Based on our review of the literature and using the UNSEEN method, this study seeks to answer particularly following research questions:

- How rare was storm Hans and how much do different return period estimations differ?
- What are the future projections of the risk of compound flood drivers and how has this risk changed over time in the past in southern Norway?

- What is the relative importance of forced climate change vs. internal variability for past, present and future compound flood drivers in Norway?

By answering these questions, the study aims to provide practical guidance on when and why each of these approaches is most appropriate for societal preparedness, and to clarify how the choice of method influences the rarity of the event. Furthermore, it should contribute to a better understanding of the complex relationship between different compound hazard factors and how they can shape the flood risk respectively.

2 Literature Review

2.1 Storm Hans: formation and impact

Storm Hans developed from the merging of two low-pressure systems over southern Scandinavia, establishing a persistent south-easterly advection of warm and humid air. Ascending at the Scandinavian mountain range, it produced several days of orographically enhanced, quasi-stationary precipitation over south-eastern Norway [Norwegian Water Resources and Energy Directorate (NVE), 2023; Mužić et al., 2026]. This circulation is not typical for the region, as usually south-eastern Norway lies in the lee of the mountain range towards the prevailing Atlantic westerlies. Heaviest summer rainfall is typically short-lived and of convective nature [Poschlod et al., 2020; Dyrørdal et al., 2016]. Parts of eastern Norway received up to 200 mm of rain within three days, accumulations that had not been observed in the region for more than a century [Granerød et al., 2023]. The event induced widespread shallow landslides and debris flows in the affected catchment areas [Rüther et al., 2024]. It also caused severe flooding along the major watercourses of eastern Norway, including the Glomma, the country’s largest river [Norwegian Water Resources and Energy Directorate (NVE), 2023].

However, estimates of the rarity of Hans differ markedly depending on the diagnostic and the record used. The event was classified as a 1-in-100-year event on average, primarily due to the unusual persistence of the south-easterly flow rather than record-breaking instantaneous intensities [Norwegian Water Resources and Energy Directorate (NVE), 2023; Mužić et al., 2026]. From a hydrological perspective, the severity of storm Hans was further amplified by an exceptionally wet July that had largely drained the storage capacity of soils and groundwater, meaning that an unusually large fraction of the rainfall directly transformed into runoff [Norwegian Water Resources and Energy Directorate (NVE), 2023]. Mužić et al. [2026] re-simulated the synoptic situation of storm Hans under different warming scenarios in order to examine how the impact of such weather events changes in a warmer climate. They find that event-total precipitation and the maximum hourly intensity increase by a factor of 1.5-2 of the Clausius–Clapeyron rate of about 7 % per degree of warming [Mužić et al., 2026].

2.2 Compound flood drivers in Norway

Compound events are combinations of multiple drivers that contribute to societal or environmental risk. Their defining feature is that the contributing variables don’t need to be extreme individually for the combination to have a significant impact [Zscheischler et al., 2018; Bevacqua et al., 2021]. In Zscheischler et al. [2020]’s definition, rain falling on an existing snowpack or saturated soil constitutes a *preconditioned* event, where an antecedent catchment state that is not hazardous itself can amplify the response to a following hazard. Which precondition matters the most depends on the region and climate zone. Berghuijs et al. [2019] analyzed the separate relative importance of extreme precipitation, soil-moisture excess and snowmelt across 3 777 European catchments. They found that annual maximum precipitation was the least important driver Europe-wide (mean $\alpha_p = 0.23$) and snowmelt and rain-on-snow (mean $\alpha_s = 0.29$) were the dominant mechanism specifically across Scandinavia and north-eastern Europe. Therefore, most annual floods in the study region are not caused by the annual rainfall peak. In southern Norway, this finding is supported by event-based evidence of two of the largest floods in

1995 and 2013 resulting from a combination of rainfall and snowmelt [Krøgli et al., 2018; Poschlod et al., 2020].

However, the relative importance of these mechanisms is not constant. Vormoor et al. [2016] found that rainfall had become a more important flood-generating process, while the role of snowmelt had declined. They also discovered that snowmelt-driven floods now occurred roughly 5–10 days earlier per decade. Beyond the magnitude of the drivers, their temporal alignment is also important: Han et al. [2025] show that combining observed precipitation events with observed antecedent catchment states systematically produces floods more severe than any on record. They also demonstrate that displacing an event by only one month can substantially amplify it. Storm Hans occurred in August, well outside the snowmelt season. The same amount of precipitation falling in May would have encountered both wet soils and an active snowpack [Han et al., 2025].

2.2.1 Rain-on-Snow events

Rain-on-snow (ROS) events require the simultaneous occurrence of liquid precipitation, sufficient snowpack and snow cover. Taken in isolation, none of these factors need to be extreme though [Pall et al., 2019]. Using the 1 km seNorge dataset [Lussana et al., 2019] over the period 1957–2016, Pall et al. [2019] produced the first nationwide ROS climatology for Norway. ROS events cluster in winter and spring along the south-western coast and the western flank of the mountain region, where Atlantic cyclone systems deliver rain onto a winter snowpack. They then shift to the high-elevation mountain and northern regions in late spring and early summer, where snow persists into the warm season. ROS frequency decreases at low and coastal elevations with climate change, which is consistent with a shortening snow season in a warming climate. In contrast, it increases at higher elevations where the snowpack persists, while a growing proportion of precipitation falls as rain. This elevation-dependent dipole is the central signature of ROS change in Norway and recurs in projections [Pall et al., 2019]. ROS intensity, defined as the sum of rainfall and snowmelt, increased in every season with climate change. This is driven by rainfall in autumn and winter, and by snowmelt in spring and summer [Mooney and Li, 2021]. Importantly for flood risk, ROS-related high flows (95th percentile) increase in both frequency and magnitude in winter and autumn. Towards the end of the century, the snowpack constraint becomes dominant. Poschlod et al. [2020] define ROS as the joint exceedance of the 98th percentile of liquid precipitation and snowmelt in the 50-member CRCM5-LE over southern Norway. They predict a 48% fall in ROS frequency under RCP8.5, driven by a 59% reduction in mean surface snow amount. Meanwhile, the probability of heavy summer rainfall on saturated soil is set to increase by 38%. Together, these studies imply that the flood regime will become progressively less governed by snowmelt and increasingly triggered by heavy rainfall and saturation excess.

Nevertheless, Poschlod et al. [2020] report that the occurrence probabilities of both compound event types vary greatly between individual ensemble members. For summer saturation-excess events, these probabilities range from 46 % below to 115 % above the ensemble median. The authors attribute this spread to internal variability in the timing of the co-occurrence rather than to the intensity of the drivers. However, the relative contributions of forced change and internal variability have not been explicitly quantified. This gap motivates the research questions posed in Section 1.3.

3 Data

For this study, the Reanalysis datasets ERA5 and seNorge as well as the 2 climate models CESM2-LE and GFDL-SPEAR-MED-LE were utilised.

3.1 ERA5 Reanalysis Dataset

The ERA5 reanalysis dataset provided by the European Center for Medium-Range Weather Forecasts (ECMWF) serves as one data source in this study. ERA5 spans the period from 1 January 1940 to the present and is produced by using the Integrated Forecasting System (IFS) cycle CY41R2 [Hersbach et al., 2020]. It employs a hybrid four-dimensional variational data assimilation scheme (4D-Var) with a 12-hour assimilation window [Courtier et al., 1994]. The ERA5 reanalysis dataset assimilates a wide range of satellite and in-situ observations.

ERA5 provides global atmospheric data at an hourly temporal resolution and a horizontal resolution of either $0.5^\circ \times 0.5^\circ$ (~ 62 km) or $0.25^\circ \times 0.25^\circ$ (~ 31 km), with 137 vertical levels extending from the Earth's surface to 0.01 hPa. The dataset includes surface and upper air variables such as precipitation and soil moisture, variables that are used extensively in this study [Soci et al., 2024].

As a reanalysis product, ERA5 represents the best physically consistent estimate of the atmospheric state, combining observations with model output [Hersbach et al., 2020]. However, uncertainties remain, particularly for small-scale processes such as convective precipitation linked to extratropical cyclones, due to limitations in model parameterizations [Urraca et al., 2018]. Although ERA5 represents past weather and climate conditions coherently and comprehensively, it should not be treated as equivalent to direct observations [Wernli and Schwierz, 2006].

3.2 Senorge Reanalysis Dataset

The second reanalysis dataset utilised in this study is called "seNorge", a high-resolution (1x1 km), gridded climatological and hydrological database covering the Norwegian mainland. The model provides daily historical and projected data for key weather and climate parameters based on observational data that is interpolated. [Lussana et al., 2019]

3.3 CESM2-LE Climate Model

CESM2-LE is a 100-member large ensemble of climate change projections conducted with the Community Atmosphere Model version 6 (CAM6) over 1850–2100. The resolution accounts for 1.25° in longitude and 0.9° in latitude over 32 vertical levels up to a height of 40km. The model uses historical (1850-2014) and future (2015-2100) climate projections, using the high-forcing SSP3-7.0 climate scenario. CESM2-LE is initialised from four years (1231, 1251, 1281, 1301) within a long-running preindustrial control simulation with a stable and small global energy imbalance, representative for a 1850-like starting state of the atmosphere. The four starting years represent the different phases of the Atlantic Meridional Overturning Circulation (AMOC) during its maximum, minimum, decreasing and increasing phases. [Rodgers et al., 2021]

3.4 GFDL-SPEAR-MED-LE Climate Model

GFDL-SPEAR-MED-LE is a 30-member large ensemble of climate change projections conducted with the SPEAR (Seamless System for Prediction and EArth System Research) modelling system over the period 1921–2100. The resolution accounts for approximately 0.5° (50km) horizontal grid spacing over 33 vertical levels. The model follows CMIP6 protocols, utilizing historical (1921–2014) and future (2015–2100) climate projections, typically employing the high-forcing SSP5-8.5 climate scenario. GFDL-SPEAR-MED-LE is initialized from a long-running 1850 preindustrial control simulation with independent restart files taken at 20-year intervals, starting with the year 101 and ending with the year 681 (30 ensembles). [Delworth et al., 2020]

3.5 Variables

Table 1 shows the derived variables and their respective unit and grid resolution used in this study.

Dataset	Variable	Unit	Grid
ERA5	Precipitation (tp24)	m $\rightarrow$ mm	0.25°
ERA5	Precipitation (tp24)	m $\rightarrow$ mm	0.5°
ERA5 regrided	Precipitation (tp)	mm d $^{-1}$	$0.94^\circ \times 1.25^\circ$
ERA5 regrided	Snow water equivalent (sd)	kg m $^{-2}$	$0.94^\circ \times 1.25^\circ$
ERA5 regrided	Soil water, 0–289 cm (swv1)	kg m $^{-2}$	$0.94^\circ \times 1.25^\circ$
seNorge	Precipitation (rr)	mm	1×1 km
CESM2-LE	Precipitation (PRECT)	mm d $^{-1}$	$0.94^\circ \times 1.25^\circ$
CESM2-LE	Snow water equivalent (SWE)	kg m $^{-2}$	$0.94^\circ \times 1.25^\circ$
CESM2-LE	Soil moisture (SM)	kg m $^{-2}$	$0.94^\circ \times 1.25^\circ$
GFDL-SPEAR-MED-LE	Precipitation (tp24)	mm d $^{-1}$	$0.5^\circ \times 0.625^\circ$

Table 1: Variables used in this report. All fields are daily. “ERA5 regrided” denotes ERA5 remapped onto the CESM2-LE grid. Names in monospace are the variable identifiers in the respective datasets. seNorge precipitation is archived as kg m $^{-2}$, numerically identical to mm, and accumulated 06–06 UTC.

4 Methods

The methods described here fall into three groups. Sections 4.1–4.3 establish the common basis on which every subsequent analysis rests. This includes the study catchments, the aggregation of gridded fields onto them, and the definition of the three compound flood drivers. Section 4.4 addresses the estimation of the rarity of Storm Hans. Sections 4.5 and 4.6 evaluate whether the large ensemble CESM2-LE climate model reproduces the observed climate closely enough for the conclusions drawn from them to be meaningful, and Section 4.7 examines the drivers jointly rather than isolated to enable a compound hazard analysis.

4.1 Study region and event definition

The analysis focuses on the five catchments most affected by Storm Hans in south-eastern Norway: Bergheim, Hønnfoss, Losna, Drammen and Glomma. All fields were cropped to a common domain of 3° – 16° E, 56.5° – 66° N covering all of these catchments. The event is treated as a multi-day precipitation extreme and is characterised by the annual maximum of the N -day catchment-averaged precipitation. Although the identical procedure was applied to all catchments, for the sake of clarity, all results in this report are presented for the Drammen+Glomma catchment only. Furthermore, a season-specific analysis was performed in this study, with the months March, April, May and June representing the extended spring season especially of importance for the snowmelt and precipitation compound flood driver.

4.2 Catchment Aggregation

As the catchments are small relative to the grid spacing of the climate models, all catchment-scale quantities represent area-fraction-weighted averages. Catchment boundaries were taken from the NEVINA and REGINE polygon datasets (GeoJSON, EPSG:4326) and dissolved into a single geometry per catchment. In the case of the combined Drammen–Glomma catchment, the union of the two polygons was formed so that overlapping areas are only counted once. Catchment geometries and grid-cell footprints, defined as rectangles spanning half a grid spacing around each cell centre, were reprojected to EPSG:25833 (UTM zone 33N) before any area computation, since at $\approx 60^{\circ}$ N a degree of longitude spans roughly half the distance of a degree of latitude. Each cell i was then assigned a weight $w_i \in [0, 1]$ equal to the fraction of its area lying inside the catchment. Weights were constructed in this way for the ERA5 0.25° , CESM2-LE and GFDL-SPEAR-MED-LE grids.

4.3 Derived Variables for Spatial Analysis

The three compound flood drivers multi-day precipitation, snowmelt, and soil moisture are considered throughout this study. None of them is used in the form archived in Table 1 as precipitation and soil moisture are multi-day transformations and snowmelt is not archived at all. ERA5 precipitation was hereby converted from m to mm, whereas seNorge precipitation (kg m^{-2}) is numerically identical to mm already. Both large ensembles archive mm d^{-1} for precipitation, so no conversion is required. Snow water equivalent (SWE) and

soil moisture were retained in kg m^{-2} . Rolling operations were applied to the full record before the analysis period was imposed, so that every window is complete.

Precipitation accumulated over an N -day window is the trailing rolling sum

$$\text{Precip}_N(t) = \sum_{k=0}^{N-1} P(t-k), \quad (1)$$

assigned to the last day of the window.

Snowmelt is archived in neither CESM2-LE nor ERA5 and was derived from the decrease in SWE across the same window,

$$\text{Snowmelt}_N(t) = \max\left(0, -\left[\text{SWE}(t) - \text{SWE}(t - (N-1))\right]\right), \quad N \geq 2, \quad (2)$$

which yields a non-negative melt variable in kg m^{-2} and assigns the value zero to days of net snow accumulation, so that accumulation does not contaminate the seasonal cycle. Snowmelt was computed per grid cell prior to any catchment aggregation, preventing accumulation in one cell from compensating melt in another.

Soil moisture is a state variable rather than an accumulation and was therefore averaged over the same window,

$$\text{SoilMoisture}_N(t) = \frac{1}{N} \sum_{k=0}^{N-1} S(t-k), \quad (3)$$

representing the mean wetness of the soil column from 0–289 cm.

4.4 Return Period Estimation

The magnitude of Storm Hans is placed in a climatological context by fitting a generalised extreme value (GEV) distribution to the annual maxima of the catchment-averaged N -day precipitation. For each catchment and dataset, the largest N -day catchment mean within each calendar year is extracted, and the GEV distribution

$$F(x) = \exp\left\{-\left[1 + \xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi}\right\}, \quad 1 + \xi\left(\frac{x-\mu}{\sigma}\right) > 0, \quad (4)$$

with location μ , scale $\sigma > 0$ and shape ξ is fitted by the maximum likelihood method. Return levels follow from the inverse distribution function, $z_T = F^{-1}(1 - 1/T)$, and the return period of an event of magnitude x from the inverse exceedance probability,

$$T(x) = \frac{1}{1 - F(x)}. \quad (5)$$

The fit is assessed against the empirical distribution using Weibull plotting positions, in which rank i of a sample of size n sorted in descending order is assigned $T_i = (n+1)/i$.

Most affected regions recorded their maximum two-day accumulation either on 7–8 or on 8–9 August 2023, so the magnitude of Storm Hans is defined as the annual maximum of 2023 rather than as a fixed pair of dates. This makes the definition independent of the accumulation window, which poses no issue, as storm Hans is accountable for the maximum for 2023 no matter the window. Second, Storm Hans is retained in the sample when fitting: removing the largest value would bias the fitted upper tail low and thereby inflate the estimated return period.

4.5 Model Bias Evaluation

Conclusions drawn from the large ensembles are only meaningful if the ensembles reproduce the observed distribution of the flood drivers over the study area. To evaluate this, the same catchment-averaged time series used in Section 4.4, over 1985–2024, is computed. In order to do so, the annual maximum sample evaluation on the one hand contains one value per year and member, whereas on the other hand, the daily sample evaluates every N -day rolling value and represents the full distribution, concatenated across members for the ensembles. Sample sizes therefore differ between products, which must be borne in mind when comparing the resulting distributions. Distribution shape is compared using a Gaussian kernel density estimate evaluated on a grid common to all products, complemented by summary statistics of the same samples.

As these comparisons characterise the distributions as a whole, they do not reveal how the bias varies in the upper tail of the dataset. A percentile–percentile mapping is used for this purpose, where for each percentile $p = 1, 2, \dots, 99$ of the climate-model sample, its respective reanalysis sample percentile is investigated. Placing both quantities on a common percentile scale allows products with very different sample sizes and record lengths to be compared on equal terms.

4.6 Spatial Bias and Significance Testing

Beyond the catchment-integrated evaluation, CESM2-LE is compared with ERA5 across the study domain over 1995–2024. ERA5 (0.5°) was first remapped onto the CESM2-LE grid, so that differences reflect the model rather than a resolution mismatch. For each grid cell, the median and the 90th percentile of each N -day field were computed over the analysis period. These statistics were evaluated in two ways for CESM2-LE: on the one hand, they were pooled over all members and time steps, and on the other hand separately per member. The overall statistic describes the modelled climate whereas the per-member statistics form a distribution which approximates the forced state and quantifies internal variability. The test is therefore formulated generally as the position of a reference value within the ensemble distribution. The reference is judged to lie outside the range of internal variability when ERA5 either falls outside the $[100 - p_{\text{upper}}]$ percentile range of the ensemble, indicating that the ensemble is systematically wetter than the reanalysis, or $[p_{\text{upper}}]$, implying that the ensemble is systematically drier than the reanalysis. The two-sided thresholds $p_{\text{upper}} = 95$ ($p \leq 0.10$) and $p_{\text{upper}} = 98$ ($p \leq 0.04$) were applied, with $n = 100$ members for precipitation and $n = 90$ for snow water equivalent (SWE) and soil moisture.

Two limitations follow from this construction. As significance is determined by counting members, attainable p -values are discretised in steps of $1/n$, so distinctions finer than roughly 0.011 cannot be resolved for the 90-member fields. As a second limitation, the current approach does not apply a False Discovery Rate (FDR) correction (see for example Wilks [2016]) to account for the multiple comparisons made across grid points. As a consequence, the nominal significance thresholds may result in an elevated false discovery rate over the domain as a whole. These limitations must be considered when interpreting the spatial extent and reliability of the statistically significant patterns.

The comparison was repeated separately for each season (DJF, MAM, JJA, SON), and additionally an extended spring-season analysis was applied (MAMJ) to account for the

months with highest snowmelt. To avoid leaving the first $N - 1$ days of each season undefined, the N -day rolling window is applied to the full record first and the result subset by season afterwards, with windows spanning a season boundary assigned to the season of their closing day.

4.7 Joint Distribution of Compound Drivers

This analysis is limited to the CESM2-LE, as it is the only ensemble in this study archiving precipitation, snow water equivalent and soil moisture simultaneously. Land-surface output variables are available for 90 of its 100 members, and the analysis is restricted to these 90 for all three variables so that every pair of values originates from the same simulation. For each member and each catchment, daily fields were reduced to catchment means (Section 4.2) before the appropriate N -day operator of Section 4.3 was applied.

We quantify with a compound severity threshold

$$\text{thr} = \frac{x}{x_{\max}} + \frac{y}{y_{\max}}, \quad (6)$$

where x and y are the two drivers of the pair and $x_{\max}$, $y_{\max}$ their sample maxima over a fixed reference selection. A pair is classified as a compound extreme when its score reaches a fixed cut-off thr^* , i.e. when $\text{thr} \geq \text{thr}^*$, which corresponds geometrically to lying above a straight line in the (x, y) graph. This enables us to have both variables contributing equally towards the compound hazard which is the case for an event very extreme in one driver alone as well as by an event only moderately severe in both drivers but severe in combination. Because the denominators are sample maxima, the score is defined relative to that reference selection rather than against an absolute threshold.

4.8 Trends and Internal Variability

To address the second and third research questions (Section 1.3), the compound criterion of Section 4.7 is evaluated over time, so that the forced change in the frequency of compound extremes can be evaluated within the ensemble members. Furthermore, the internal variability can be evaluated. No declustering is applied, so one day counts as one event and a single storm can therefore contribute to more than one day.

Two complementary analysis are derived from the same computation and therefore share the settings listed in Table 2. The first shows the evolution of the event frequency of every ensemble member against the ensemble mean, which approximates the forced climate response and hence addresses the second research question. The second analysis depicts the signal-to-noise ratio, which relates the forced response to the spread across members and hence addresses the third research question. The settings in Table 2 are configurable and can be adjusted to the desired analysis conditions. The catchment, the driver variables (precipitation, snowmelt or soil moisture) and the event duration N define which compound hazard is examined. Following the methodology of Section 4.7, the extreme compound events are defined by the compound-threshold thr^* and the reference period over which the maxima $x_{\max}$ and $y_{\max}$ are selected. This criterion remains fixed in order to end up with a straight line in the (x, y) graph utilised for the entire analysis window. Exceedance days are then counted per member and calendar year and aggregated over rolling windows of a chosen length and step, yielding for each member a rate in events per year which is

plotted at the window centre. The months retained and the number of ensemble members M included are also specified in Table 2.

Parameter	Meaning	Value Example
Catchment / pair	compound hazard analysed	Drammen–Glomma, precip. + snowmelt
N	event accumulation window	2 days
thr*	compound-severity threshold (Eq. 6)	0.7
Reference period	fixed $x_{\max}$, $y_{\max}$	1995–2024, all members, MAMJ
Analysis period	time-range of full record analysed	1920–2034
L , Δ	rolling-window length, step	10 years, 1 year/season
Season	analyzed months	MAMJ (March–June)
M	ensemble size	90 members
Percentile estimator	empirical or grouped (Eq. 8)	grouped

Table 2: Settings of the trend and internal-variability analysis.

For each window the M member rates are visualized by their ensemble mean $\bar{f} = M^{-1} \sum_m f_m$ together with the interquartile range (25th–75th percentile) and the full ensemble range or the empirical 2.5th–97.5th percentile range. The latter is obtained from the M member rates sorted in ascending order by linear interpolation between the two compound driver statistics so that it is less sensitive to a single outlying member. Because all members share the same external forcing and differ only in their initial conditions, the spread within a window is a direct estimate of the internal variability present at that time. It is quantified by the sample standard deviation σ across members and related to the forced signal (ensemble mean) by the dimensionless signal-to-noise ratio

$$S/N = \frac{\bar{f}}{\sigma}, \quad \sigma = \sqrt{\frac{1}{M-1} \sum_{m=1}^M (f_m - \bar{f})^2}. \quad (7)$$

The mean serves as the central estimate throughout the dataset where a member rate is a count divided by L and can therefore only represent a multiple of $w = 1/L$. On the other hand, $\bar{f}$ resolves $1/(ML)$ and therefore no longer keeps this discretization. As empirical percentiles land exactly on that grid as well, grouped percentiles can be applied in order to remove this artefact. This is ensured by treating each attainable rate as the centre of a bin of width w and interpolating through the tied block,

$$\hat{q}_p = b_{\text{lo}} + \frac{pM/100 - F}{f} w_b, \quad (8)$$

with b_{lo} and w_b the lower end and width of the bin and F and f the member counts below and within that bin. The zero bin is kept at half width, $[0, w/2)$, because the event frequency cannot be negative.

5 Results

5.1 Return-Period Estimation of Storm Hans

Figure 1 illustrates the return period for Storm Hans, based on the 1985–2024 ERA5 (0.5° resolution) reanalysis time series (see top of Figure 1) as well as according to the CESM2-LE climate model for the 1985–2024 period, using the reference precipitation value from the Senorge reanalysis tool (see bottom of Figure 1). ERA5 as a reanalysis tool also contains a time series of 2-day precipitation for the same period. The respective annual maxima are marked as blue points and fitted to a black line (GEV fit), whereas the red point represents Storm Hans with the corresponding precipitation value. The event is estimated to occur approximately once every 140 years (ERA5) and once every 3,100 years (CESM2-LE), revealing significant differences between the two reanalysis/modelling tools. Figure 1 therefore clearly indicates that it is very difficult to estimate the exact return period of such an event.

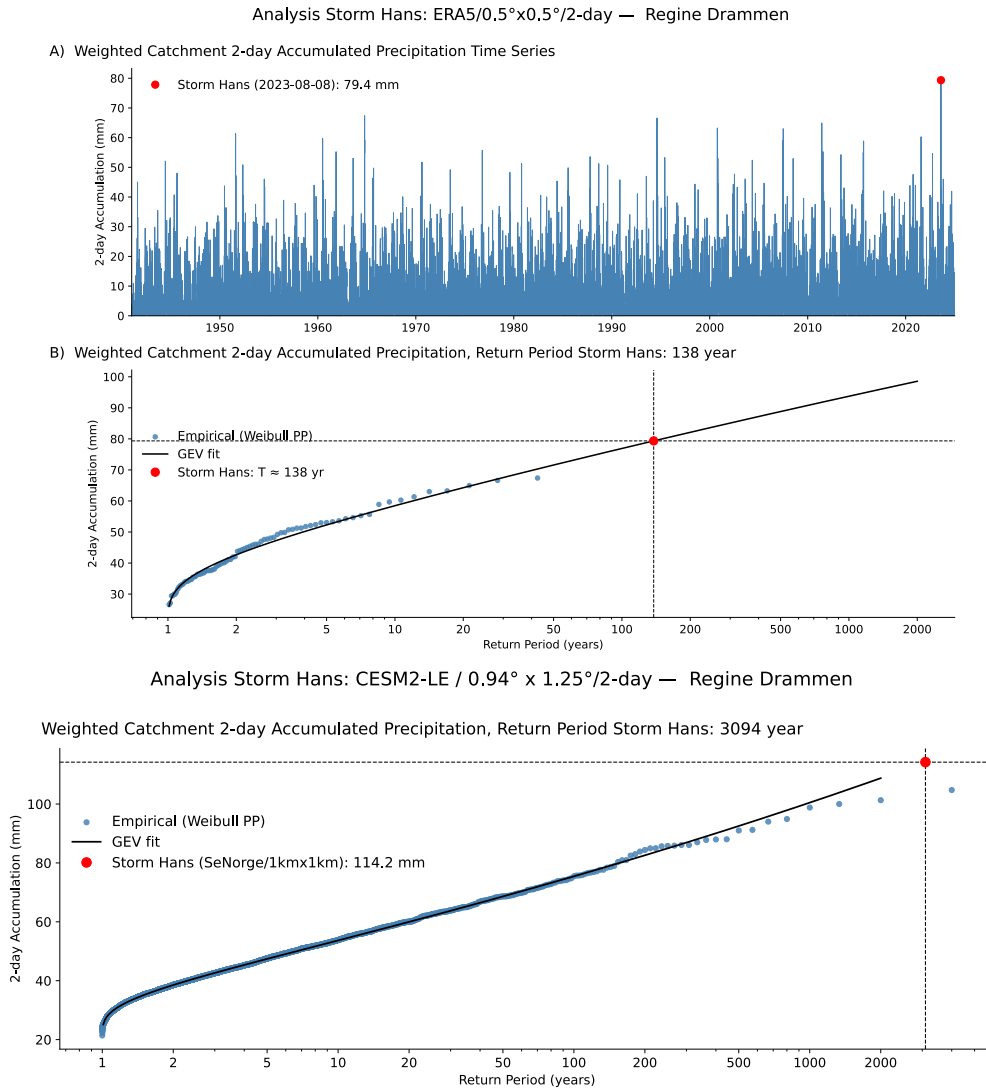


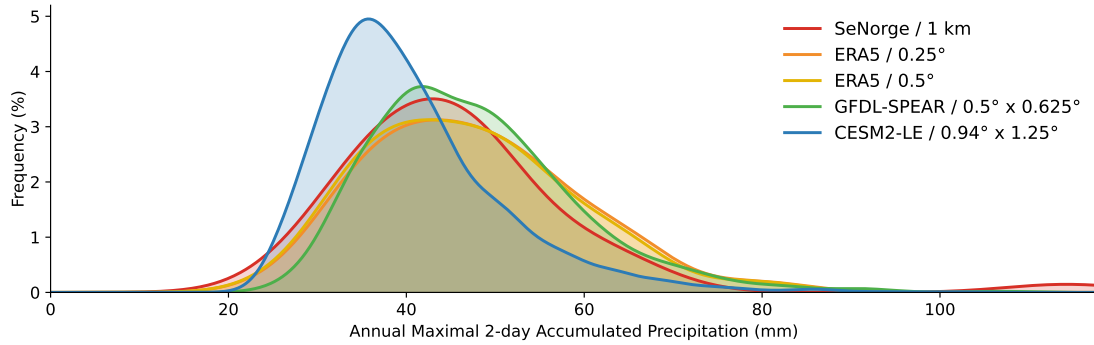
Figure 1: Exemplary Storm Hans Return Period Estimations for Catchment Drammen

5.2 Model Bias Evaluation CESM2-LE

Figure 2A) illustrates the frequency distribution of values for annual maximum 2-day precipitation between 1985 and 2024 across various models and reanalysis tools. Apart from CESM2-LE which has a significantly narrower range of values around the maximum than the reanalysis tools and GFDL-SPEAR, all models account for a similar distribution of annual 2-day precipitation maxima. Figure 2B) presents this information in the form of box plots, which allow the distributions to be further quantified. The interquartile range predominantly extends from 38–53 mm, with a median of around 44 mm. Only the CESM2-LE climate model has a median below 40 mm and a lower interquartile range. Furthermore, Figure 2B) demonstrates that the number of outliers is significantly higher for the two climate models than for the reanalysis tools, since the mean number of simulations is increased by considering multiple ensemble members instead of a single reanalysis.

Distribution of Annual Maximal 2-day Accumulated Precipitation — Regine Drammen

A) Frequency Distribution



B) Boxplot Distribution

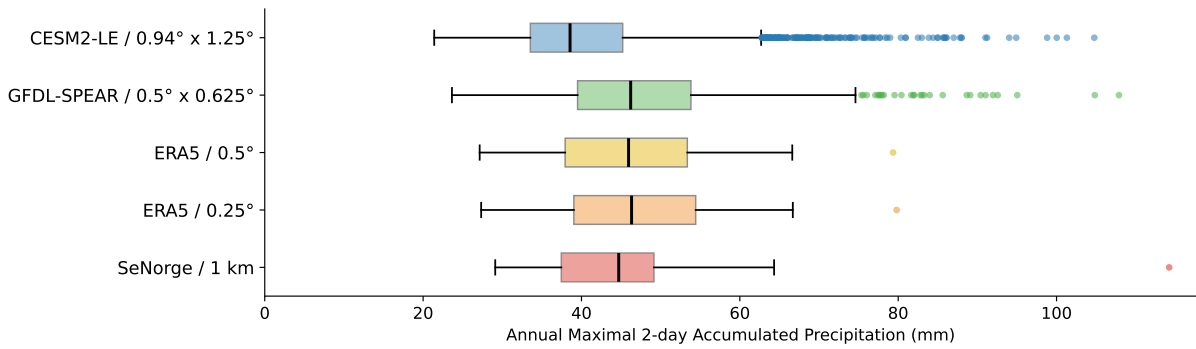


Figure 2: Distribution of Annual Maximal 2-day Accumulated Precipitation (1985-2024) for different Model Types for Catchment Drammen

5.3 Compound Flood Risk Analysis

5.3.1 Spring (MAMJ) Analysis of Precipitation & Snowmelt

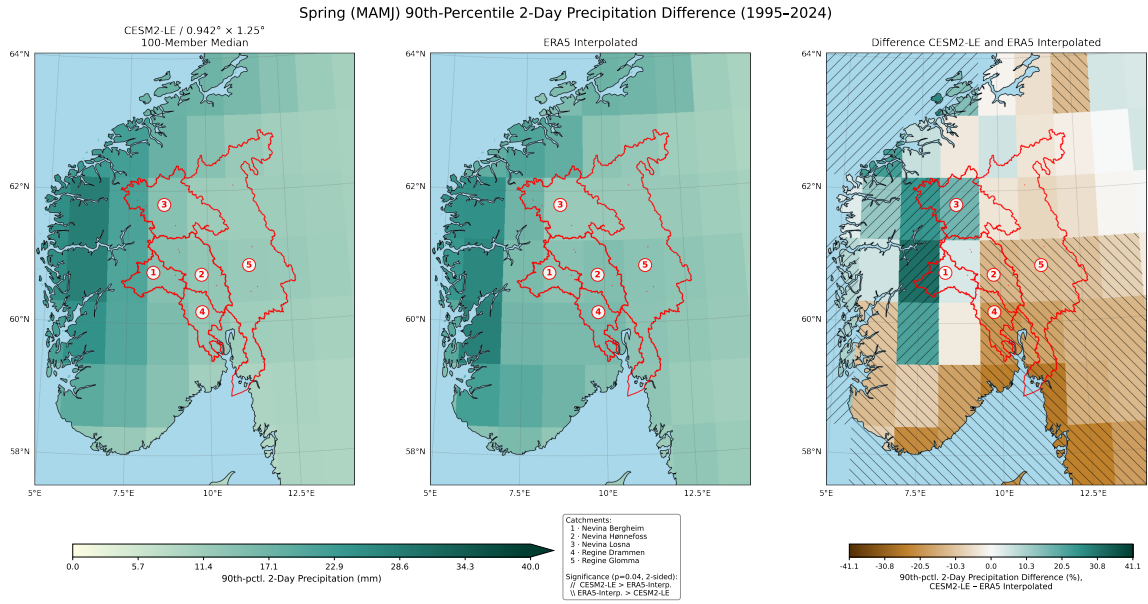


Figure 3: 2-Day Spring (MAMJ) 90th-Percentile Precipitation Difference between CESM2-LE and ERA5 from 1995–2024

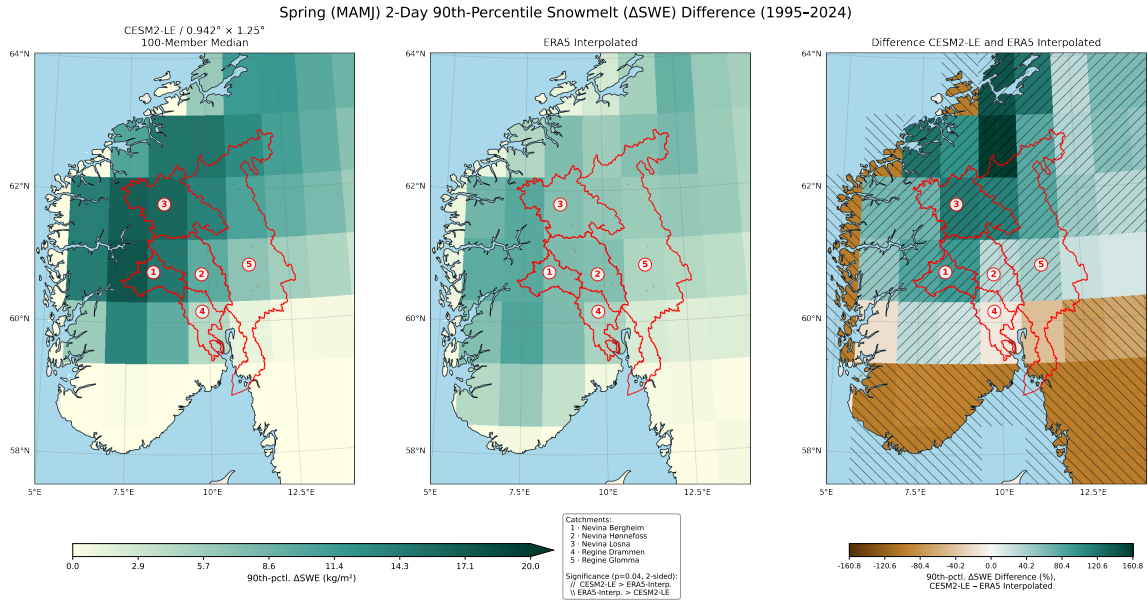


Figure 4: 2-Day Spring (MAMJ) 90th-Percentile Snowmelt (SWE) Difference between CESM2-LE and ERA5 from 1995–2024

Figure 3 depicts the 90th percentile of 2-day precipitation from 1995 to 2024 in spring (MAMJ), with the five catchment areas most severely affected by Storm Hans delineated by red boundaries. The left-hand panel indicates the median for all 100 CESM2-LE climate model members, the middle panel the ERA5 reanalysis values (0.5° regridded onto CESM2-LE), and the right-hand panel the difference between both models. Additionally, grid cells with a statistically significant difference are highlighted by hatching. Moreover, Figure 4 illustrates the same information, but for the 90th percentile of 2-day snowmelt

from 1995 to 2024 in spring (MAMJ) rather than precipitation. Figure 3 shows that parts of the catchment areas with higher topography exhibit significantly higher precipitation values in CESM2-LE than in ERA5. Conversely, the 90th percentile 2-day precipitation in spring in CESM2-LE is significantly underestimated compared to ERA5 in the flatter southern areas of the catchments. Notably, CESM2-LE accounts for significantly higher snowmelt than ERA5 across most of the catchment areas. Only in the southernmost part of the Glomma catchment, ERA5 values present significantly higher snowmelt rates (see Figure 4).

5.3.2 Joint Distribution of Precipitation & Snowmelt

Figure 5 illustrates the joint distribution of 2-day snowmelt and precipitation across the Drammen + Glomma catchment area of the CESM-LE climate model for the period 1995–2024, with data points coloured according to season. High snowmelt rates occur particularly in late spring and early summer (as indicated by the green data points), whereas high precipitation coupled with low snowmelt mainly occurs in midsummer and autumn (as shown by the blue and violet data points). A threshold line with a reference value of 0.75 further highlights areas with a high degree of overlap between both snowmelt and precipitation, representative of the compound hazard events of interest (see Figure 5).

Joint Distribution of 2-Day Precipitation Sum and 2-Day Snowmelt (SWE Difference), 1995–2024

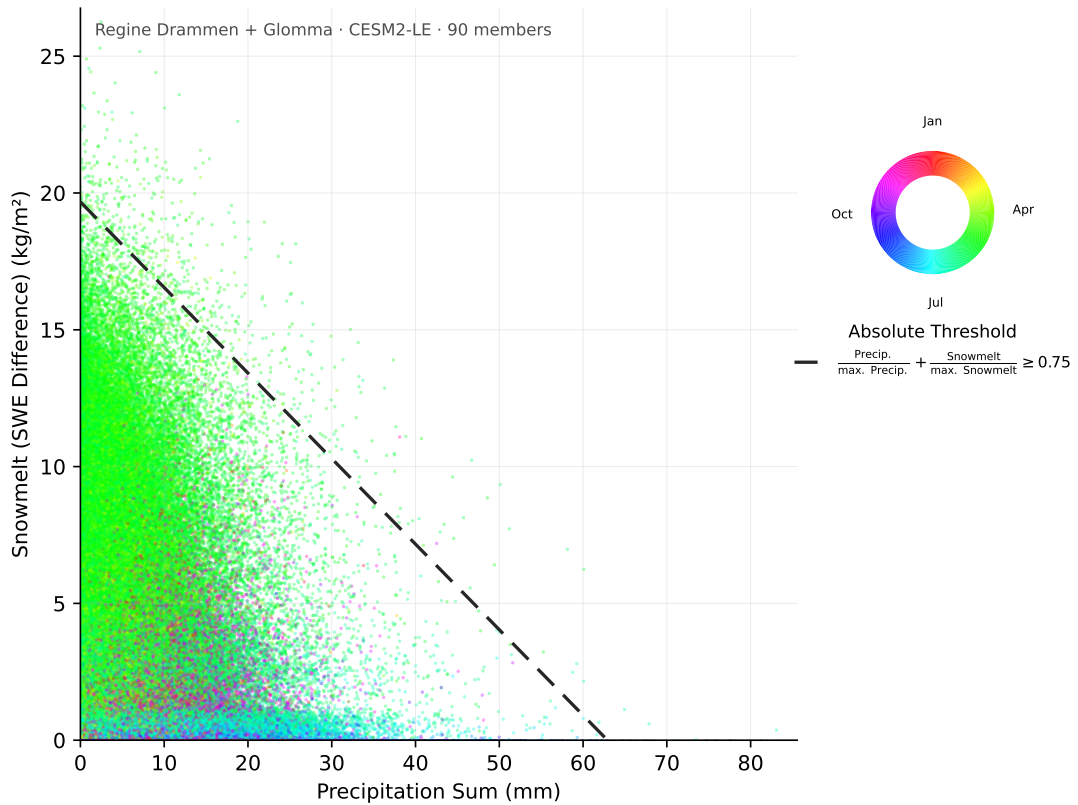


Figure 5: Joint Distribution of 2-Day Precipitation Sum and 2-Day Snowmelt for Catchment Drammen+Glomma, 1995–2024

5.3.3 Temporal Evolution of the Compound Hazard Frequency

Figures 6 and 7 depict the evolution of the spring (MAMJ) compound hazard of 2-day precipitation and 2-day snowmelt over the combined Drammen–Glomma catchment for the 90 CESM2-LE members over 1920–2034. A day is counted as a compound extreme when the threshold reaches $\text{thr} \geq 0.7$ (Figure 6) or $\text{thr} \geq 0.9$ (Figure 7) for the reference period 1995–2024. Rates are averaged within a 10-year centred rolling window and plotted at the window centre. The solid line represents the ensemble mean, the blue shading the interquartile range (Equation 8) and the dotted lines the 2.5th and 97.5th member-percentiles.

At a threshold of $\text{thr} \geq 0.7$ (see Figure 6) the ensemble mean decreases from about 0.65 events per spring season in the window centred on 1925 to about 0.38 at the end of the record, a reduction of roughly 40 %. The spread across members decreases as well, but less than the mean. In every window the interquartile range spans roughly a factor of two around the ensemble mean, and the 97.5th percentile lies between two and three times above it.

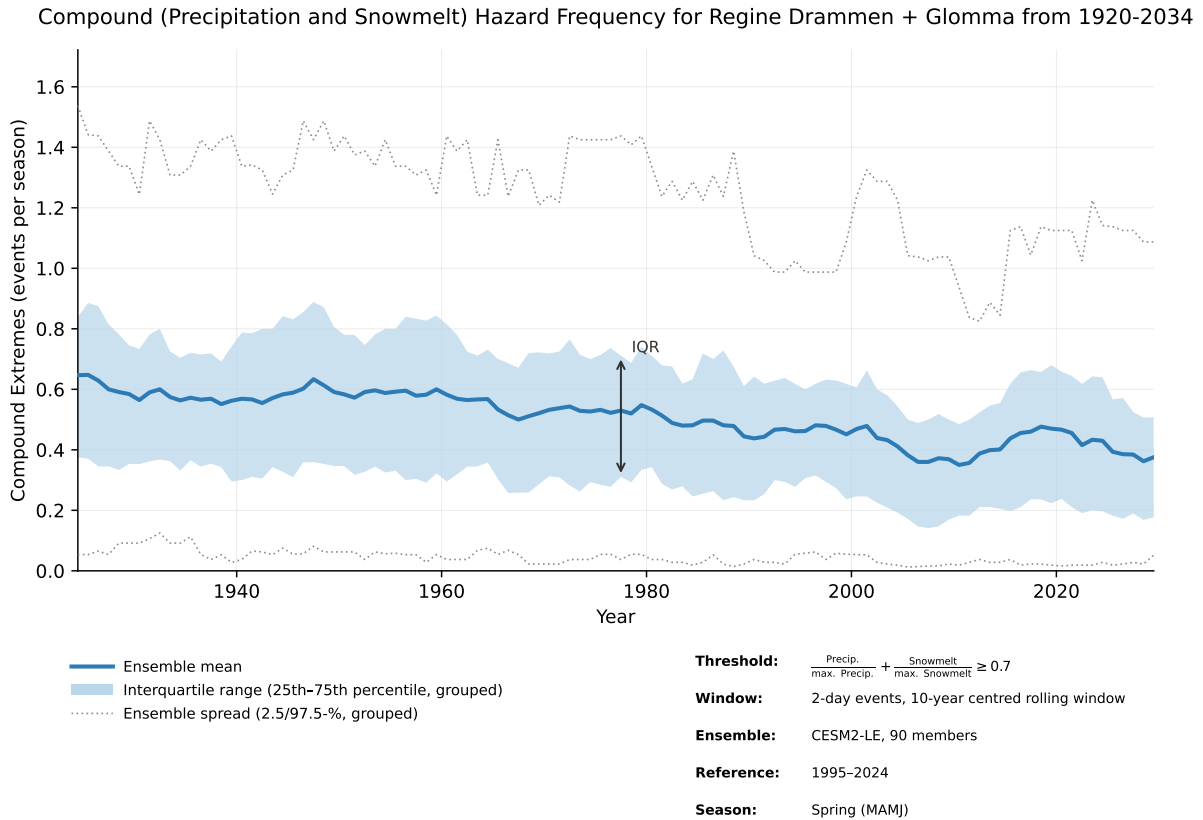


Figure 6: Evolution of 2-Day Spring (MAMJ) Compound (Precipitation & Snowmelt) Hazard Frequency for Catchment Drammen+Glomma from 1920-2034 (10 year window, threshold of 0.7, reference period 1995-2024)

At the higher threshold $\text{thr} \geq 0.9$ (see Figure 7) the frequency is more than an order of magnitude lower and no comparable trend is present. The 25th percentile is almost constant at about 0.017 events per season and the 75th percentile as well as the 97.5th percentile fluctuates without a clear trend. The 2.5th percentile is indistinguishable from zero in every window of the record, meaning that in each decade the lowest members

simulate no compound extreme at all.

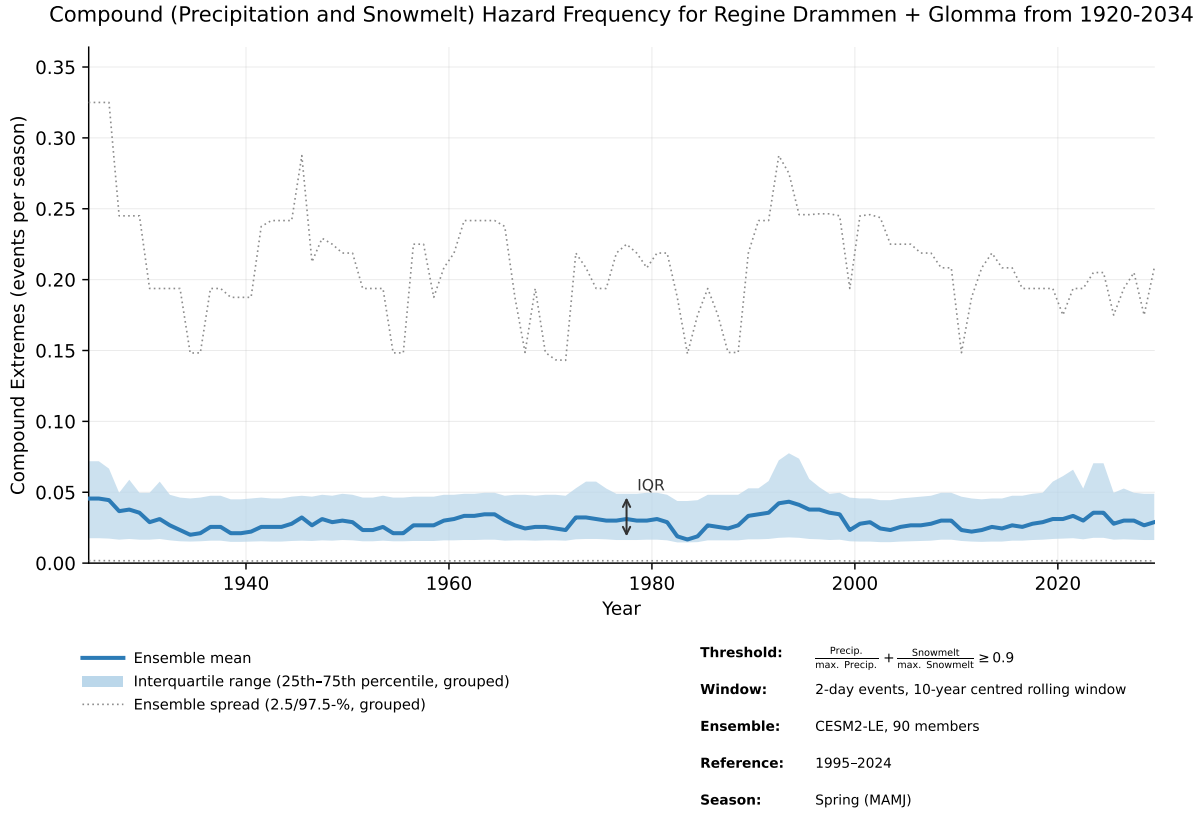


Figure 7: Evolution of 2-Day Spring (MAMJ) Compound (Precipitation & Snowmelt) Hazard Frequency for Catchment Drammen+Glomma from 1920-2034 (10 year window, threshold of 0.9, reference period 1995-2024)

5.3.4 Signal-to-Noise Ratio of the Compound Flood Driver

Figures 8 and 9 show the signal-to-noise ratio of Equation 7, defined as the ensemble mean divided by the ensemble member standard deviation of the same 10-year window. A value of 1.0 corresponds to a standard deviation equal to the mean, and a value of 0.5 to a standard deviation twice the mean.

With a threshold of $\text{thr} \geq 0.7$ (see Figure 8) the ratio remains between 1.21 and 2.06 over the analysis record, with a long-term average of about 1.66. Until roughly 1990 it doesn't show a trend, but then falls to about 1.4 for the rest of the analysis period. At a threshold of $\text{thr} \geq 0.9$ (see Figure 9) the ratio lies below unity in every single window. It varies between 0.37 and 0.61 around an average of 0.48, so that the inter-member standard deviation is approximately twice the ensemble mean throughout the analysis period. No systematic trend can be observed between 1920-2034.

Signal-to-noise-ratio Compound (Precipitation and Snowmelt) Hazard for Regine Drammen + Glomma from 1920-2034

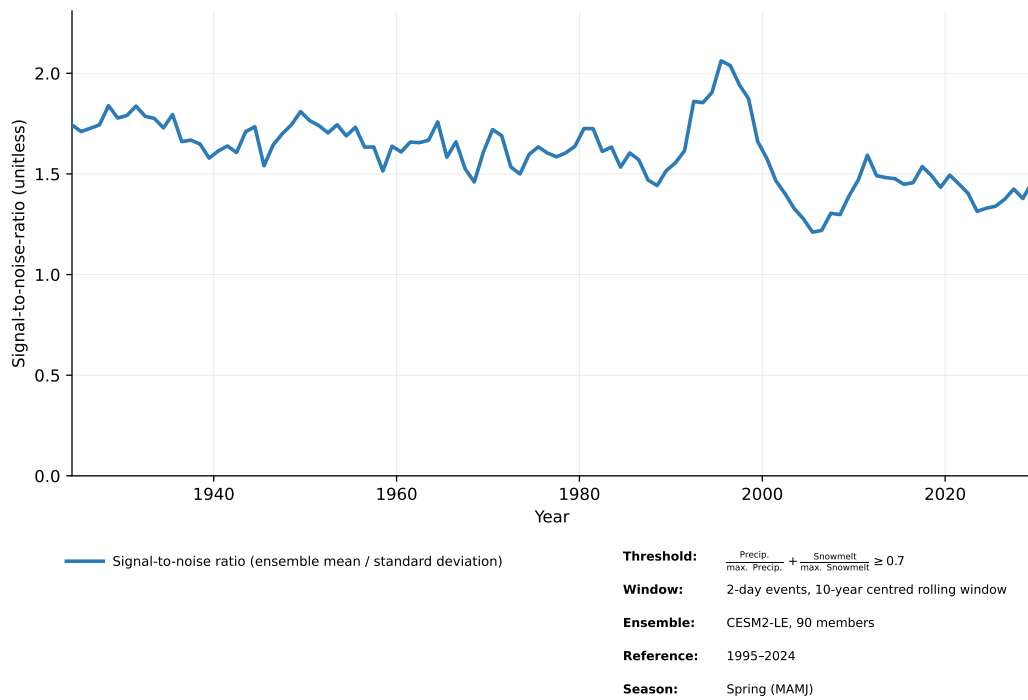


Figure 8: Evolution of 2-Day Spring (MAMJ) Compound (Precipitation & Snowmelt) Hazard Signal-to-Noise Ratio for Catchment Drammen+Glomma from 1920-2034 (10 year window, threshold of 0.7, reference period 1995-2024)

Signal-to-noise-ratio Compound (Precipitation and Snowmelt) Hazard for Regine Drammen + Glomma from 1920-2034

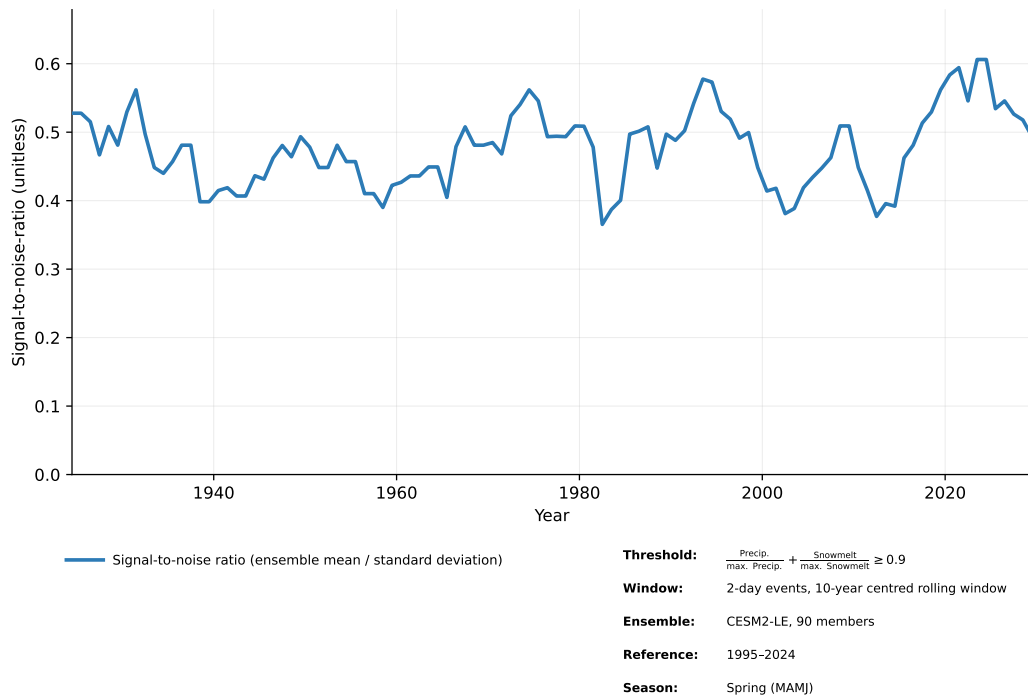


Figure 9: Evolution of 2-Day Spring (MAMJ) Compound (Precipitation & Snowmelt) Hazard Signal-to-Noise Ratio for Catchment Drammen+Glomma from 1920-2034 (10 year window, threshold of 0.9, reference period 1995-2024)

6 Discussion

6.1 Evolution of Compound Snowmelt + Precipitation Frequency

As all 90 members share the same external forcing and only differ in their initial conditions, their mean isolates the forced climate response. At the threshold $\text{thr} \geq 0.7$ this response is clear, as the frequency of spring compound days falls by approximately 40 % across the record. This accounts for a change eight times larger than the noise of the ensemble mean itself ($\sigma/\sqrt{M}$). This magnitude is close to the 48 % of reduction in rain-on-snow frequency stated out by Poschlod et al. [2020] under the RCP8.5 scenario and it is consistent with the shortening snow season and declining snowmelt contribution documented by Pall et al. [2019] and Vormoor et al. [2016]. The fact that the rainfall side of the compound is not diminishing underlines that the decline is driven by the snowpack rather than the precipitation term.

The threshold $\text{thr} \geq 0.9$ shows no forced trend at all. Snow loss with climate change reduces days with moderate snowmelt and precipitation while leaving the severe compound driver tail almost unchanged. Mooney and Li [2021] determine that rain-on-snow intensity rises in every season even if the frequency falls. Thus, an increasing rainfall term compensates a declining snowmelt term in the regime where both account for large values. Pall et al. [2019] describe an elevation-dependent dipole where rain-on-snow decreases at low and coastal elevations while it increases where snowpack persists. In a catchment spanning both bands (as Drammen–Glomma does) this leads to a net loss of moderate events with no clear change in the severe tail.

6.2 Discussion Internal Variability

For the threshold $\text{thr} \geq 0.7$ the inter-member standard deviation remains between roughly half and one ensemble mean throughout 1920–2034 whereas for the threshold $\text{thr} \geq 0.9$ it never falls below the mean and is typically about twice as large. The forced change at the moderate threshold is of the order of one inter-member standard deviation, so it is resolved only because 90 realisations share identical forcing and would not be identifiable in any single one. At the severe threshold the upper members exceed the ensemble mean by roughly a factor of seven in a given decade while the lower members simulate no such event at all under the same forcing.

The ratio is itself a noisy quantity: it swings on 5–15 year timescales by about a quarter of its long-term level at $\text{thr} \geq 0.7$ and by more than a factor of 1.6 at $\text{thr} \geq 0.9$, without any corresponding change in forcing and without a systematic trend at either threshold. Estimating the balance between signal and noise from one decade or from a single observational record is therefore unreliable, and a quiet decade constitutes almost no evidence that the compound drivers have declined. The drivers have a well-defined forced response, most clearly the declining snowmelt term, but whether they coincide within a two-day window in a given spring remains largely a matter of chance, as Poschlod et al. [2020] could report. Storm Hans could be explained accordingly, since an event of that extremeness is fully compatible with a declining forced signal when that signal is not the dominant term on planning timescales.

6.3 Limitations

There are some limitations in this study that need to be accounted for. First of all, the threshold is applied to a certain reference period (here: 1995-2024) so all results are relative to that reference period rather than to an absolute compound driver level. Furthermore, no declustering is applied, so a multi-day storm can potentially contribute to several event days and the data used in this analysis represents day and not storm counts. The spatial significance test applies no False Discovery Rate correction [Wilks, 2016], so the extent of the significant areas is likely overstated. Additionally, the current resolution of snowmelt doesn't coincide with the one from precipitation and soil moisture, as the mean and sum across 2 days would be representative of a 48-hour temporal resolution, consistent with an SWE difference of the N+2 day - N-day rather than N+1 day - N-day. Finally, the compound analysis rests on a single multi-ensemble climate model (CESM2-LE) meaning that model-specific biases cannot be separated from a robust signal.

6.4 Future Research

Validation against observed runoff: Additionally taking into account actual in-situ rain-gauge or river observations for the joint-distribution analysis as in Section 5.3.2 would potentially allow us to estimate if the applied methodology relates compound events to gauged discharge for Drammen and Glomma in a hydrological meaningful way. This is underlined by previous studies such as Berghuijs et al. [2019] and Vormoor et al. [2016] which attributed flood generation from discharge rather than from meteorological fields.

Restriction to liquid precipitation: Total precipitation was used throughout this study, yet only rainfall drives rain-on-snow events. A separation of liquid and solid precipitation and only taking into account rainfall would be beneficial for that.

Refinement of the event definition: The current analysis takes into account days where one driver alone carries the threshold exceedance, so a warm period producing intense snowmelt without precipitation is counted identically to a compound driver. Since rain-on-snow requires liquid precipitation, snowpack and snow cover simultaneously [Pall et al., 2019] and snowmelt alone has less potential of inducing floods than rainfall has, cutting the threshold to a certain precipitation minimum value required for the analysis might be useful.

Seasonal shift: According to the results obtained by Vormoor et al. [2016], snowmelt floods are shifting 5–10 days earlier per decade. Limiting the analysis on MAMJ might lead to omittance of an analysis of a potential seasonal shift of such compound events, suggesting an extendance of the analysis on other months.

Rain on antecedent wet soil: Storm Hans' severity partially can be described from soils and groundwater already saturated by an exceptionally wet July [Norwegian Water Resources and Energy Directorate (NVE), 2023]. This precipitation and soil moisture compound flood driver is projected to become more frequent by 38 % under RCP8.5 [Poschlod et al., 2020]. Other studies highlight that a 7-year rainfall event on saturated soil can generate a 100-year flood while a 200-year event on dry soil can potentially generate only a 15-year one [Kim et al., 2019], justifying to look deeper into this type of compound extreme for southern Norway.

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